

Week 5 Deliverable: Mobility, Wheel Torque, Power, and Motor Selection

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1 Week 5 Objective

The purpose of Week 5 is to evaluate whether the lunar regolith mover rover can drive while carrying its full regolith payload. The Week 4 CAD baseline fixed the main rover dimensions, payload size, wheel diameter, chassis size, and mass estimate. Week 5 uses that baseline to calculate the required traction force, wheel torque, wheel speed, and power for flat terrain and several slope angles.

2 Baseline Rover Parameters

The Week 4 design baseline is summarized in Table 1.

Table 1: Week 4 Rover Baseline Used for Week 5 Mobility Calculations

Parameter	Symbol	Value
Dry rover mass	m_{dry}	650 kg
Regolith payload mass	$m_{payload}$	340 kg
Loaded rover mass	m	990 kg
Lunar gravity	g	1.62 m/s ²
Dump bed inside length	L_{bed}	1.20 m
Dump bed inside width	W_{bed}	0.75 m
Dump bed inside height	H_{bed}	0.35 m
Wheel diameter	D	0.50 m
Wheel radius	r	0.25 m
Number of wheels	n	6
Number of powered wheels	$n_{powered}$	6
Baseline rover speed	v	0.20 m/s
Rolling resistance coefficient	C_{rr}	0.10
Design factor	D.F.	2.0

3 Dump Bed Volume and Payload Density Check

The internal dump bed volume is calculated from:

$$V_{bed} = L_{bed}W_{bed}H_{bed} \quad (1)$$

Substituting the Week 4 dimensions:

$$V_{bed} = (1.20)(0.75)(0.35) \quad (2)$$

$$V_{bed} = 0.315 \text{ m}^3 \quad (3)$$

The effective bulk density implied by a 340 kg payload is:

$$\rho_{regolith,eff} = \frac{m_{payload}}{V_{bed}} \quad (4)$$

$$\rho_{regolith,eff} = \frac{340}{0.315} \quad (5)$$

$$\rho_{regolith,eff} = 1079 \text{ kg/m}^3 \quad (6)$$

Therefore, the selected dump bed volume and payload mass imply an effective loose regolith bulk density of approximately:

$$\rho_{regolith,eff} \approx 1080 \text{ kg/m}^3 \quad (7)$$

This is acceptable for a conceptual design estimate because the payload and dump bed volume are internally consistent.

4 Loaded Lunar Weight

Although the rover has a loaded mass of 990 kg, the weight on the Moon is much lower than it would be on Earth. The loaded lunar weight is:

$$W = mg \quad (8)$$

$$W = (990)(1.62) \quad (9)$$

$$W = 1603.8 \text{ N} \quad (10)$$

Assuming the load is evenly distributed across all six wheels, the approximate normal load per wheel is:

$$N_{wheel} = \frac{W}{6} \quad (11)$$

$$N_{wheel} = \frac{1603.8}{6} \quad (12)$$

$$N_{wheel} = 267.3 \text{ N} \quad (13)$$

5 Mobility Force Model

The required traction force is estimated using rolling resistance and slope resistance:

$$F_{total} = mgC_{rr} + mg \sin(\theta) \quad (14)$$

where:

- F_{total} = total required traction force,
- m = loaded rover mass,
- g = lunar gravity,
- C_{rr} = rolling resistance coefficient,
- θ = terrain slope angle.

The rolling resistance force is:

$$F_{rolling} = mgC_{rr} \quad (15)$$

$$F_{rolling} = (990)(1.62)(0.10) \quad (16)$$

$$F_{rolling} = 160.4 \text{ N} \quad (17)$$

The slope force is:

$$F_{slope} = mg \sin(\theta) \quad (18)$$

The total force is then:

$$F_{total} = F_{rolling} + F_{slope} \quad (19)$$

6 Slope Performance Force Results

Table 2 shows the calculated traction force required for different slope angles.

Table 2: Required Traction Force Versus Slope Angle

Slope Angle	Rolling Force	Slope Force	Total Required Force
0°	160.4 N	0.0 N	160.4 N
5°	160.4 N	139.8 N	300.2 N
10°	160.4 N	278.5 N	438.9 N
15°	160.4 N	415.1 N	575.5 N
20°	160.4 N	548.5 N	708.9 N

The main design cases are 10° and 15°. The 20° case is included as a stress case but is not selected as the primary design slope.

7 Wheel Torque Calculation

The total torque required at the wheels is:

$$T_{total} = F_{total}r \quad (20)$$

where:

- T_{total} = total wheel torque required,
- F_{total} = total required traction force,
- r = wheel radius.

For a six-wheel-drive rover, the torque per wheel is:

$$T_{wheel} = \frac{T_{total}}{6} \quad (21)$$

A design factor of 2.0 is applied to account for uncertainty in terrain, loose regolith, startup resistance, drivetrain losses, and possible wheel sinkage:

$$T_{design} = 2T_{wheel} \quad (22)$$

Table 3: Wheel Torque Requirement Versus Slope Angle

Slope	Total Force	Total Torque	Torque/Wheel	Torque/Wheel with 2x D.F.
0°	160.4 N	40.1 N m	6.7 N m	13.4 N m
5°	300.2 N	75.0 N m	12.5 N m	25.0 N m
10°	438.9 N	109.7 N m	18.3 N m	36.6 N m
15°	575.5 N	143.9 N m	24.0 N m	48.0 N m
20°	708.9 N	177.2 N m	29.5 N m	59.1 N m

Based on Table 3, the recommended wheel motor output torque is:

$$T_{motor,output} \approx 40 \text{ N m to } 50 \text{ N m} \quad (23)$$

This torque is the required output torque after the gearbox, not the raw motor shaft torque.

8 Power Calculation

The ideal traction power is calculated using:

$$P = F_{total}v \quad (24)$$

where:

- P = ideal traction power,
- F_{total} = required traction force,
- v = rover speed.

The baseline speed is:

$$v = 0.20 \text{ m/s} \quad (25)$$

Table 4: Ideal Traction Power Versus Slope Angle

Slope	Total Force	Ideal Traction Power	Power with 2x D.F.
0°	160.4 N	32.1 W	64.2 W
5°	300.2 N	60.0 W	120.1 W
10°	438.9 N	87.8 W	175.6 W
15°	575.5 N	115.1 W	230.2 W
20°	708.9 N	141.8 W	283.6 W

The ideal traction power is relatively low because lunar gravity is low and the assumed rover speed is slow. However, the actual mobility power system should be sized higher to

include motor inefficiency, gearbox losses, startup torque, wheel slip, control electronics, and uneven terrain.

Therefore, the recommended total mobility power sizing is:

$$P_{mobility,design} \approx 200 \text{ W to } 300 \text{ W} \quad (26)$$

9 Wheel Speed Calculation

The angular speed of the wheel is:

$$\omega = \frac{v}{r} \quad (27)$$

$$\omega = \frac{0.20}{0.25} \quad (28)$$

$$\omega = 0.80 \text{ rad/s} \quad (29)$$

Converting to revolutions per minute:

$$RPM = \omega \frac{60}{2\pi} \quad (30)$$

$$RPM = 0.80 \frac{60}{2\pi} \quad (31)$$

$$RPM = 7.64 \quad (32)$$

Therefore, the wheel output speed requirement is approximately:

$$RPM_{wheel} \approx 7.6 \text{ rpm} \quad (33)$$

The selected wheel motor and gearbox should therefore provide approximately:

$$40 \text{ N m to } 50 \text{ N m} \quad \text{at approximately } 8 \text{ rpm} \quad (34)$$

10 Rolling Resistance Sensitivity Check

Because lunar regolith conditions can vary, the rolling resistance coefficient was also varied from 0.05 to 0.20 for the 10° slope case. This checks whether the selected torque range is still reasonable if the surface is more difficult than expected.

Table 5: Sensitivity of Torque and Power to Rolling Resistance at 10° Slope

C_{rr}	Total Force	Torque/Wheel	Torque/Wheel with 2x D.F.	Power with 2x D.F.
0.05	358.7 N	14.9 N m	29.9 N m	143.5 W
0.10	438.9 N	18.3 N m	36.6 N m	175.6 W
0.15	519.1 N	21.6 N m	43.3 N m	207.6 W
0.20	599.3 N	25.0 N m	49.9 N m	239.7 W

This sensitivity check supports the recommended design torque of 40 N m to 50 N m per wheel after gearing.

11 Traction and Slip Check

The previous calculations estimate the force required to move the rover. However, the rover must also have enough wheel-regolith traction to actually provide that force without excessive slipping. A simple traction check can be performed using:

$$F_{available} = \mu W \quad (35)$$

where:

- $F_{available}$ = maximum available traction force,
- μ = assumed wheel-regolith traction coefficient,
- W = loaded lunar weight of the rover.

For this preliminary analysis, the wheel-regolith traction coefficient is assumed to be:

$$\mu = 0.50 \quad (36)$$

The loaded lunar weight was previously calculated as:

$$W = 1603.8 \text{ N} \quad (37)$$

Therefore, the available traction force is:

$$F_{available} = (0.50)(1603.8) \quad (38)$$

$$F_{available} = 801.9 \text{ N} \quad (39)$$

The traction condition is:

$$F_{available} > F_{required} \quad (40)$$

Table 6: Traction and Slip Check

Slope	Required Force	Available Traction Force	Pass/Fail
0°	160.4 N	801.9 N	Pass
5°	300.2 N	801.9 N	Pass
10°	438.9 N	801.9 N	Pass
15°	575.5 N	801.9 N	Pass
20°	708.9 N	801.9 N	Pass, lower margin

A traction margin can also be calculated as:

$$\text{Traction Margin} = \frac{F_{\text{available}}}{F_{\text{required}}} \quad (41)$$

Table 7: Traction Margin Versus Slope Angle

Slope	Required Force	Traction Margin
0°	160.4 N	5.00
5°	300.2 N	2.67
10°	438.9 N	1.83
15°	575.5 N	1.39
20°	708.9 N	1.13

The traction check shows that the rover has enough available wheel-regolith traction for the evaluated slope cases when $\mu = 0.50$ is assumed. The 15° case passes with a margin of 1.39, while the 20° stress case still passes but has a much smaller margin of 1.13. This means the rover is reasonable for mild slopes, but steeper slopes or lower-traction regolith could create slip concerns.

12 Drivetrain and Gearbox Efficiency

The torque and power values calculated earlier are wheel output requirements. In reality, the motor and battery must provide more power than the ideal wheel power because energy is lost in the gearbox, bearings, motor controller, and drivetrain.

The drivetrain efficiency is assumed to be:

$$\eta_{\text{drive}} = 0.70 \text{ to } 0.85 \quad (42)$$

The required motor input power is:

$$P_{\text{motor,input}} = \frac{P_{\text{wheel}}}{\eta_{\text{drive}}} \quad (43)$$

Using the 15° slope case:

$$P_{wheel} = 115.1 \text{ W} \quad (44)$$

For a conservative drivetrain efficiency of 70 percent:

$$P_{motor,input} = \frac{115.1}{0.70} \quad (45)$$

$$P_{motor,input} = 164.4 \text{ W} \quad (46)$$

For a drivetrain efficiency of 85 percent:

$$P_{motor,input} = \frac{115.1}{0.85} \quad (47)$$

$$P_{motor,input} = 135.4 \text{ W} \quad (48)$$

Therefore, the motor and battery must provide more than the ideal wheel power. The ideal 115.1 W traction power at a 15° slope becomes approximately 135 W to 164 W before adding design factor or non-mobility loads.

Table 8: Motor Input Power Including Drivetrain Efficiency

Slope	Ideal Wheel Power	Input Power at $\eta = 0.85$	Input Power at $\eta = 0.70$
0°	32.1 W	37.8 W	45.9 W
5°	60.0 W	70.6 W	85.8 W
10°	87.8 W	103.3 W	125.4 W
15°	115.1 W	135.4 W	164.4 W
20°	141.8 W	166.8 W	202.6 W

This supports the earlier recommendation that the mobility traction system should be sized around 200 W to 300 W for slow driving after drivetrain losses and design margin are included.

13 Gear Ratio Estimate

The calculated wheel speed is only about 7.6 rpm. A brushless DC motor will usually spin much faster than this before gear reduction, so a gearbox is required.

The gear ratio can be estimated using:

$$G = \frac{RPM_{motor}}{RPM_{wheel}} \quad (49)$$

Assuming a representative brushless DC motor speed of approximately 3000 rpm:

$$G = \frac{3000}{7.64} \quad (50)$$

$$G = 392.7 \quad (51)$$

Therefore, the estimated gear ratio is approximately:

$$G \approx 400 : 1 \quad (52)$$

If the wheel speed is rounded to 8 rpm, then:

$$G = \frac{3000}{8} \quad (53)$$

$$G = 375 : 1 \quad (54)$$

This supports the use of a high-ratio planetary or harmonic gearbox. The gearbox allows the motor to operate at a higher efficient speed while producing the low-speed, high-torque wheel output needed for lunar regolith mobility.

14 Energy Per Trip Estimate

A basic energy-per-trip estimate can be made using:

$$E_{trip} = P_{design} t_{trip} \quad (55)$$

where:

- E_{trip} = energy required per trip,
- P_{design} = design mobility power,
- t_{trip} = trip time.

Assuming a trip time of approximately 14 minutes:

$$t_{trip} = 14 \text{ min} \quad (56)$$

Convert to hours:

$$t_{trip} = \frac{14}{60} \quad (57)$$

$$t_{trip} = 0.233 \text{ hr} \quad (58)$$

Using the upper mobility power estimate:

$$P_{design} = 300 \text{ W} \quad (59)$$

The mobility energy per trip is:

$$E_{trip} = (300)(0.233) \quad (60)$$

$$E_{trip} = 69.9 \text{ Wh} \quad (61)$$

Therefore, the traction mobility energy for one trip is approximately:

$$E_{trip} \approx 70 \text{ Wh} \quad (62)$$

However, this only includes the mobility traction system. After adding sensors, communication, onboard computing, dump bed actuation, motor controller losses, and operational margin, a more realistic conceptual energy estimate is:

$$E_{trip,total} \approx 100 \text{ Wh to } 200 \text{ Wh} \quad (63)$$

This estimate can be used later for preliminary battery sizing.

15 Motor Selection Approach

At this stage, a final flight-qualified motor is not selected. Instead, a representative motor specification is created from the calculated torque, speed, and power requirements. This is appropriate for a conceptual rover design because the purpose is to size the mobility system, not to complete a final procurement-level motor selection.

The assumed motor architecture for the rover is:

Six independent brushless DC gearmotors, one per wheel, with planetary or harmonic gear reduction, encoder feedback, motor controller, and a holding brake.

The motor is sized at the wheel output, meaning after gear reduction. The design requirement is:

$$T_{output} = 40 \text{ N m to } 50 \text{ N m} \quad (64)$$

$$RPM_{output} \approx 8 \text{ rpm} \quad (65)$$

A higher peak torque should also be assumed for startup, obstacle contact, sinkage, and short-duration overloads:

$$T_{peak} \approx 80 \text{ N m to } 100 \text{ N m} \quad (66)$$

15.1 Why Brushless DC Motors Are Considered

Brushless DC motors are considered for the concept rover because they are commonly used in robotic and aerospace mechanisms where controlled torque, low maintenance, and high reliability are important. Unlike brushed DC motors, brushless motors do not use mechanical brushes for commutation. This reduces brush wear, friction, and maintenance concerns.

For a lunar rover, the main advantages are:

- no brush wear during long-duration operation,

- precise speed and torque control using electronic commutation,
- compatibility with encoder feedback,
- good efficiency for battery-powered robotic systems,
- easier integration with motor controllers,
- improved suitability for sealed drive units exposed to dust and vacuum.

The main disadvantages are:

- higher controller complexity,
- need for electronic commutation,
- thermal management concerns in vacuum,
- need for radiation-tolerant or protected electronics in a real flight system.

NASA rover examples support the use of geared electric wheel actuators. Curiosity’s wheel and steer actuator architecture used brushless DC motors with a high-ratio planetary gearbox, brake, and encoder [2]. NASA’s VIPER mobility design also used dedicated motor controllers and integrated mobility actuators for lunar surface mobility [3, 4]. Apollo’s Lunar Roving Vehicle used a different historical architecture with brushed DC traction motors and harmonic drives, which provides a useful comparison point [5, 6]. NASA also studied brushless motor concepts for the Lunar Roving Vehicle program, although the flight vehicle ultimately used brushed motors to reduce development and testing risk [7].

16 Brushless Motor Anatomy

Figure 1 displays an exploded top display of a brushless DC motor. It uses electronically controlled stator coils to create a rotating magnetic field, which interacts with permanent magnets on the rotor to produce torque. Reversing the commanded magnetic field sequence changes the direction of rotation.

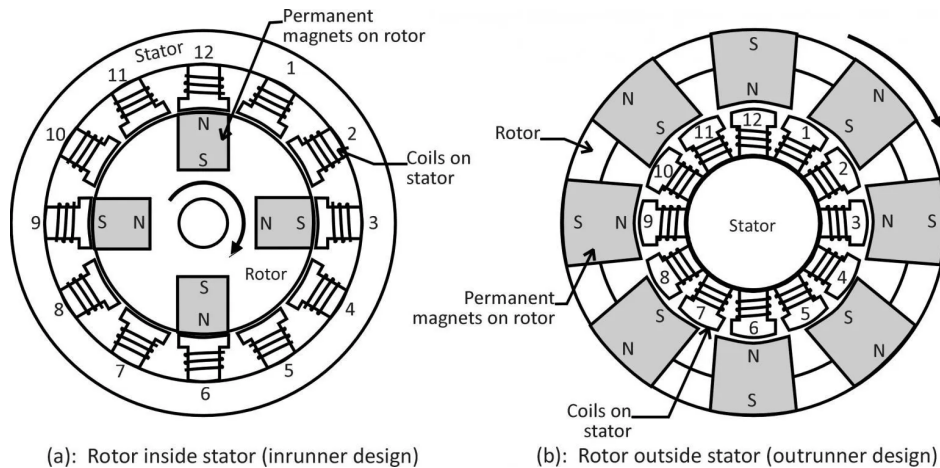


Figure 1: Brushless DC motor anatomy

17 Motor Type Comparison

Several motor options were considered for the conceptual rover. The goal is not to select a final commercial motor, but to choose a realistic motor architecture for early-stage sizing.

Table 9: Comparison of Possible Wheel Motor Types

Motor Type	Advantages	Disadvantages	Decision
Brushed DC gear-motor	Simple control, historically used in Apollo LRV traction drives	Brush wear, friction, sealing concerns, less attractive for long-duration robotic operation	Not selected as primary concept
Brushless DC gearmotor	High efficiency, no brush wear, good torque control, works well with encoders and controllers	Requires motor controller and electronic commutation	Selected for concept
Stepper motor	Simple position control and high holding ability	Lower efficiency, heat generation, less ideal for continuous traction drive	Not selected
Direct-drive hub motor	Fewer mechanical gearbox parts	Requires high-torque low-speed motor, likely heavier	Not selected for this stage
Brushless DC motor with planetary or harmonic gearbox	Good balance of torque, speed reduction, controllability, and rover heritage	Gearbox adds complexity and efficiency loss	Preferred architecture

18 NASA Rover Motor Reference Comparison

Table 10 compares the selected conceptual motor approach with NASA rover motor examples.

Table 10: NASA Rover Motor Reference Comparison

System	Motor / Actuator Architecture	Relevance to This Rover	Use in This Project
Concept regolith mover rover	Six brushless DC gear-motors, one per wheel, with encoder, brake, and planetary or harmonic gearbox	Matches calculated need for controlled low-speed torque	Selected notional design
Apollo Lunar Roving Vehicle	Four independent brushed DC traction drives with harmonic gear reduction and brakes [5, 6]	Shows that independent electric wheel drives have real lunar heritage	Historical comparison only
NASA Curiosity Rover	Brushless DC wheel/steer actuators with 1024:1 planetary gear reduction, brake, and encoder [2]	Strong reference for geared electric rover actuators	Architecture reference, not direct motor selection
NASA VIPER Rover	Integrated mobility modules with drive, steering, and suspension actuation; Moon-rover motor controllers were tested in lunar simulant terrain [3, 4]	Strong lunar rover reference for mobility actuator integration	Architecture reference
Commercial COTS BLDC gearmotor	Readily available motor, gearbox, encoder, and controller combinations	Useful for early-stage sizing and prototype-level thinking	Conceptual sizing reference only

19 Selected Notional Motor Specification

Based on the mobility calculations and NASA rover motor references, the assumed wheel motor specification is shown in Table 11.

Table 11: Recommended Notional Wheel Motor Specification

Parameter	Recommended Concept Value
Motor type	Brushless DC gearmotor
Number of drive motors	6
Motor placement	One motor per wheel
Continuous/design output torque	40 N m to 50 N m
Peak output torque	80 N m to 100 N m
Wheel output speed	Approximately 8 rpm
Gearbox type	Planetary or harmonic reduction
Feedback	Encoder recommended
Brake	Holding brake recommended
Total mobility power sizing	200 W to 300 W

20 Engineering Summary

The loaded rover mass is estimated as:

$$m = 990 \text{ kg} \quad (67)$$

Under lunar gravity, this produces a loaded rover weight of:

$$W = 1603.8 \text{ N} \quad (68)$$

Using a rolling resistance coefficient of 0.10, a wheel radius of 0.25 m, six powered wheels, and a speed of 0.20 m/s, the required traction force is:

- 160.4 N on level ground,
- 438.9 N on a 10° slope,
- 575.5 N on a 15° slope,
- 708.9 N on a 20° stress-case slope.

The calculated wheel torque requirement is:

- 6.7 N m per wheel on level ground,
- 18.3 N m per wheel on a 10° slope,
- 24.0 N m per wheel on a 15° slope,
- 29.5 N m per wheel on a 20° stress-case slope.

After applying a 2.0 design factor, the recommended wheel motor output torque is:

$$T_{output} \approx 40 \text{ N m to } 50 \text{ N m} \quad (69)$$

The traction and slip check shows that, for an assumed wheel-regolith traction coefficient of $\mu = 0.50$, the available traction force is:

$$F_{available} = 801.9 \text{ N} \quad (70)$$

This is greater than the required force for the 15° slope case:

$$801.9 \text{ N} > 575.5 \text{ N} \quad (71)$$

The 20° stress case also passes:

$$801.9 \text{ N} > 708.9 \text{ N} \quad (72)$$

but with lower margin. This means the rover concept is reasonable for mild slopes, but steeper slopes or lower-traction regolith could create slip concerns.

The calculated wheel speed is approximately:

$$RPM_{wheel} \approx 7.6 \text{ rpm} \quad (73)$$

Assuming a brushless DC motor speed of approximately 3000 rpm, the estimated gearbox ratio is:

$$G \approx 375 : 1 \text{ to } 400 : 1 \quad (74)$$

This supports the use of a high-ratio planetary or harmonic gearbox.

The ideal traction power ranges from approximately 32 W on flat ground to 115 W on a 15° slope. After drivetrain losses and design margin, the traction mobility system should be sized for approximately:

$$P_{mobility} \approx 200 \text{ W to } 300 \text{ W} \quad (75)$$

This power estimate applies only to the mobility traction system during slow driving. The total rover electrical power will be higher after sensors, communication, onboard computing, actuators, motor controllers, battery losses, and thermal control are included.

For a 14-minute trip and a 300 W mobility power estimate, the traction energy per trip is:

$$E_{trip} = 69.9 \text{ Wh} \quad (76)$$

After adding other rover electrical loads and operational margin, a more realistic conceptual energy estimate is:

$$E_{trip,total} \approx 100 \text{ Wh to } 200 \text{ Wh} \quad (77)$$

The rover baseline is mechanically reasonable for a conceptual lunar regolith mover. Each wheel should be driven by a brushless DC gearmotor capable of approximately 40 N m to 50 N m of output torque after gearing at about 8 rpm. A peak torque capability of 80 N m to 100 N m per wheel is recommended for startup, loose regolith, wheel sinkage, and obstacle negotiation. A high-ratio planetary or harmonic gearbox around 375:1 to 400:1 is reasonable for matching a high-speed BLDC motor to the low-speed rover wheel. The total mobility traction system should be sized at approximately 200 W to 300 W, while total rover electrical power will be higher.

References

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